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# ORACLE

## Academy

# Database Foundations

**3-4**

**Data Modeling Terminology and Mapping**

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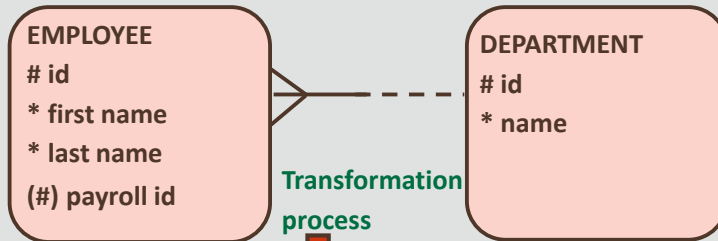
# Objectives

- This lesson covers the following objectives:
  - Apply terminology mapping between the logical and physical models
  - Understand and apply the Oracle naming conventions for tables and columns used in physical models
  - Apply the rules of relationship mapping to correctly transform relationships



# Logical To Physical Transformation: Example

Logical Model  
(ERD)



Transformation  
process

EMPLOYEES (EPE)

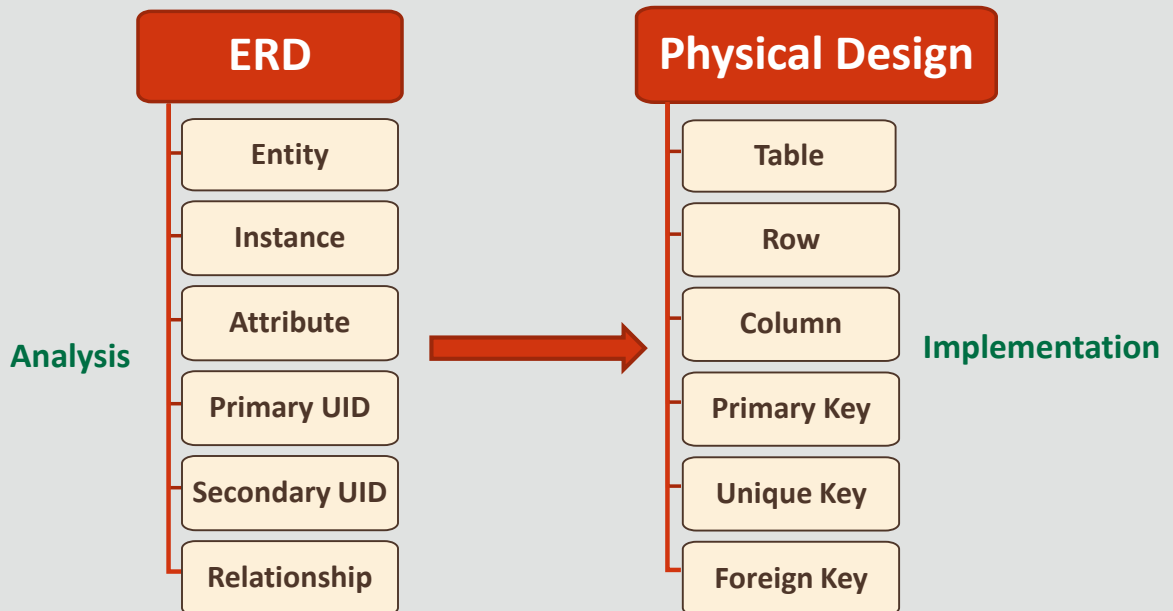
| Key Type | Optionality | Column name   |
|----------|-------------|---------------|
| pk       | *           | id            |
| uk       | *           | payroll_id    |
|          | *           | last_name     |
|          | *           | first_name    |
| fk       | *           | department_id |

Physical Implementation:  
Relational Database

DEPARTMENTS (DPT)

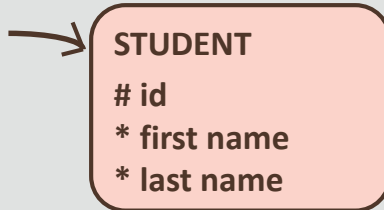
| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |
|          | *           | name        |

# Terminology Mapping



# An Entity and a Corresponding Table

Entity



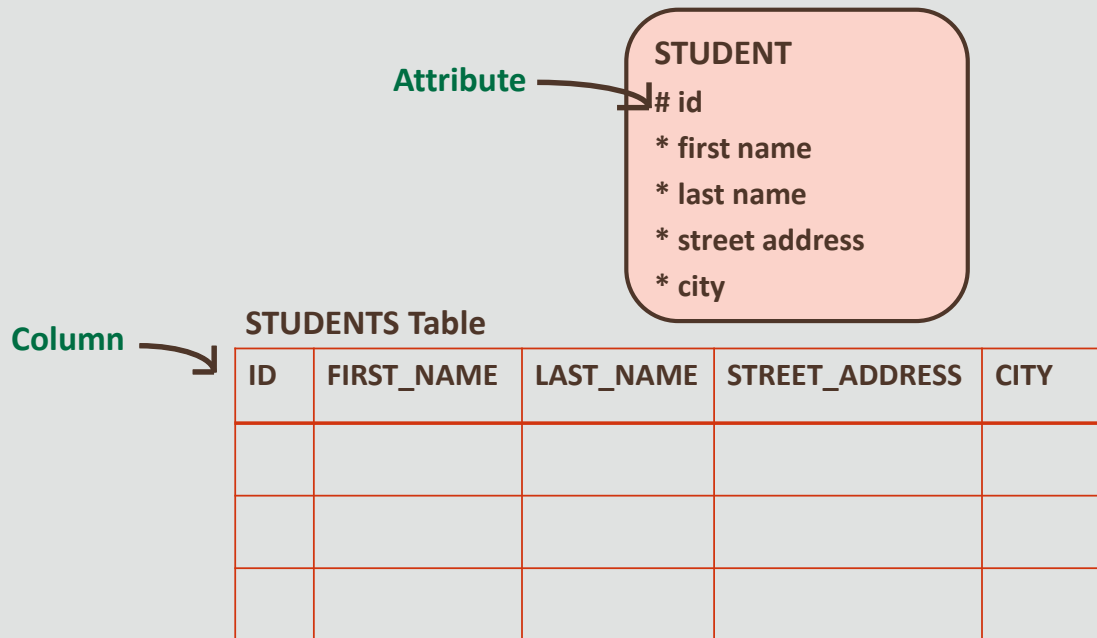
Table

**STUDENTS Table**

A rectangular table with three columns and four rows. The columns are labeled **ID**, **FIRST\_NAME**, and **LAST\_NAME**. The first row contains the column headers, and the following three rows are empty. A green arrow points from the word "Table" to the top-right corner of the table.

| ID | FIRST_NAME | LAST_NAME |
|----|------------|-----------|
|    |            |           |
|    |            |           |
|    |            |           |

# Attributes and Corresponding Columns



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Attributes are classified as one of the following:

- **Not null (mandatory):** Indicated by the asterisk ( \* ) symbol next to the attribute
- **Optional (nulls allowed):** Indicated by the o (optional) symbol next to the attribute

# An Instance and a Corresponding Row

**Entity**  
**STUDENT**

**Instance**  
**J Smith**

| ID  | FIRST_NAME | LAST_NAME | STREET_ADDRESS                     | CITY |
|-----|------------|-----------|------------------------------------|------|
| 101 | Sam        | Linkin    | 99B, Chuah Street                  | LA   |
| 102 | Neena      | Markin    | 44A, Church Street                 | NZ   |
| 103 | Rick       | Austina   | 1 <sup>st</sup> Cross, Palm Street | SA   |
| 104 | J          | Smith     | Alpha Street                       | CA   |

→ **Row**



## Table Diagram Notations

- A table diagram is additional documentation that is often used to further explain keys and columns in the physical database

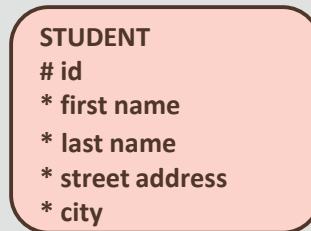
**STUDENTS** Table

| Key Type | Optionality | Column Name    |
|----------|-------------|----------------|
| pk       | *           | id             |
|          | *           | first_name     |
|          | *           | last_name      |
|          | *           | street_address |
|          | *           | city           |

- The Key Type column should contain values of "pk" for the primary key, "uk" for the unique key, or "fk" for the foreign key column. The cell is blank if the column is not a part of a key.
- The Optionality column must contain an asterisk (\*) if the column is mandatory and a lowercase "o" if it is optional. This is similar to the ERD.
- The third column is for the column name.

# Naming Conventions for Tables

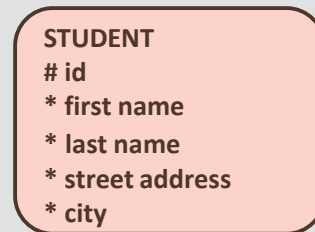
- The table name is the plural of the entity name.
  - Example: STUDENT becomes STUDENTS.
- Column names are identical to the attribute names, except that special characters and spaces are replaced with underscores



| STUDENTS |             |                |
|----------|-------------|----------------|
| Key Type | Optionality | Column name    |
| pk       | *           | id             |
|          | *           | first_name     |
|          | *           | last_name      |
|          | *           | street_address |
|          | *           | city           |

# Naming Conventions for Columns

- Column names are similar to the attribute names except that special characters and spaces are replaced with underscores
- Column names often use more abbreviations than attribute names.
  - Example: First name becomes first\_name or fname



| STUDENTS |             |                |
|----------|-------------|----------------|
| Key Type | Optionality | Column name    |
| pk       | *           | id             |
|          | *           | first_name     |
|          | *           | last_name      |
|          | *           | street_address |
|          | *           | city           |

## Table Short Names

- A unique short name for every table is useful when naming foreign key columns

PRIVATE HOME

# id

\* address

o comments

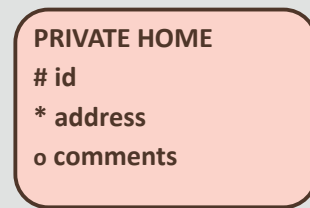


PRIVATE\_HOMES (PHE)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |
|          | *           | address     |
|          | o           | comments    |

## Table Short Names

- Create short names based on:
  - Entity names that contain more than one word
  - Entity names that contain one word but more than one syllable
  - Entity names that contain one syllable but more than one character



| PRIVATE_HOMES (PHE) |             |             |
|---------------------|-------------|-------------|
| Key Type            | Optionality | Column name |
| pk                  | *           | id          |
|                     | *           | address     |
|                     | o           | comments    |

For entity names that contain more than one word, take the:

- First character of the first word
- First character of the second word
- Last character of the last word

Example: PRIVATE HOME gets a short name of PHE.

For entity names that contain one word but more than one syllable, take the:

- First character of the first syllable
- First character of the second syllable
- Last character of the last syllable

Example: EMPLOYEE gets a short name of EPE and CLIENT gets a short name of CET.

For entity names that contain one syllable but more than one character, take the:

- First character
- Second character
- Last character

Example: FLIGHT gets a short name of FLT.

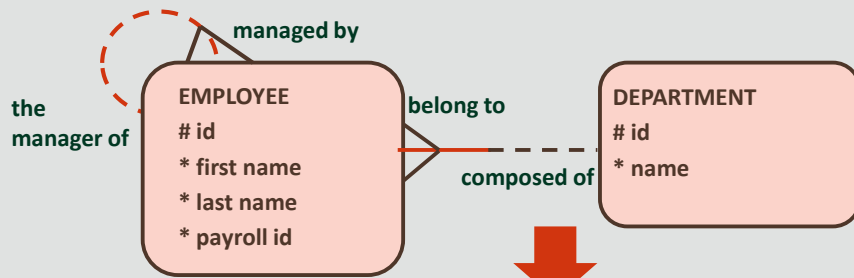
# Naming Restrictions with Oracle

- Table and column names:
  - Must start with a letter
  - Can contain up to 30 alphanumeric characters
  - Cannot contain spaces or special characters such as "!", " but "\$," "#," and "\_" are permitted
  - Cannot be “reserved words” in Oracle DB or SQL
- Table names must be unique within one user account in the Oracle database
- Column names must be unique within a table

## Relationship Mapping

- Relationships are mapped between primary keys and foreign keys to allow one table to reference another
- A relationship creates one or more foreign-key columns in the table on the many side of the relationship
- We use the short name of the table to name the foreign-key column
- In the example on the next page, the foreign-key column in the EMPLOYEES table is dpt\_id for the relationship with DEPARTMENT, and mgr\_id for the recursive relationship with itself

# Relationship Mapping Illustrated



EMPLOYEES (EPE)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |
|          | *           | first_name  |
|          | *           | last_name   |
| uk       | *           | payroll_id  |
| fk1      | *           | dpt_id      |
| fk2      | o           | mgr_id      |

DEPARTMENTS (DPT)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |
| uk       | *           | name        |

foreign key refers to

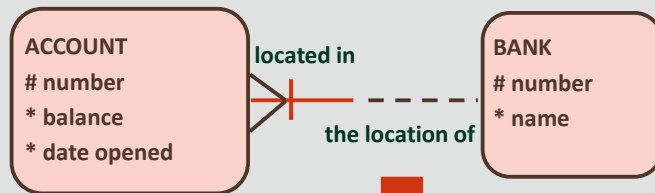
foreign key refers to



## Mapping of Barred Relationships

- A barred relationship is mapped to a foreign-key column on the many side, just like any other 1:M relationship
- In this case, the foreign-key column plays a double role because it is also part of the primary key
- In the example, bak\_number is a foreign-key column in ACCOUNTS that refers to the primary key of BANKS
- It is also part of the primary key of ACCOUNTS

# Mapping of Barred Relationships



ACCOUNTS (ACT)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | number      |
|          | *           | balance     |
|          | *           | date_opened |
| pk,fk    | *           | bak_nbr     |

BANKS (BAK)

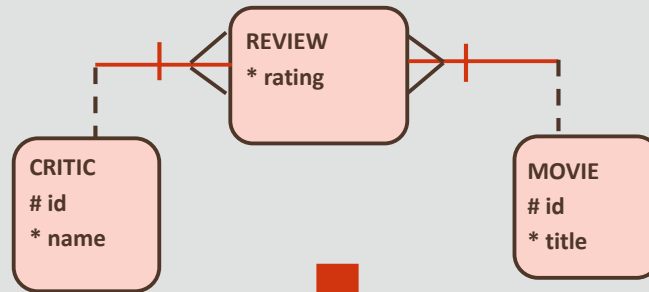
| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | number      |
|          | *           | name        |

foreign key refers to

## Mapping Many-to-Many Relationships

- A M:M relationship is resolved with an intersection entity, which maps to an intersection table
- This intersection table will contain foreign-key columns that refer to the originating tables
- In the example, REVIEWS contains all the combinations that exist between a CRITIC and a MOVIE

# Mapping Many-to-Many Relationships



REVIEWS (RVW)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk,fk1   | *           | ctc_id      |
| pk,fk2   | *           | mve_id      |
|          | *           | rating      |

CRITICS (CTC)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |
|          | *           | name        |

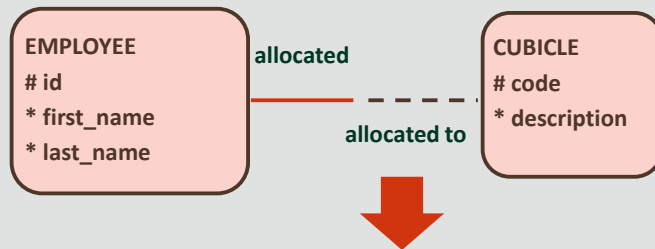
MOVIES (MVE)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |
|          | *           | title       |

## Mapping One-to-One Relationships

- When transforming a 1:1 relationship, you create a foreign key and a unique key
- All columns of this foreign key are also part of the unique key
- If the relationship is mandatory on one side, the foreign key is created in the corresponding table
- In the example, cbe\_code is the foreign-key column in EMPLOYEES that refers to the primary key of CUBICLES.
- Cbe\_code would also be unique within the EMPLOYEES table

# Mapping One-to-One Relationships



**EMPLOYEES (EPE)**

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |
|          | *           | first_name  |
|          | *           | last_name   |
| fk,uk    | *           | cbe_code    |

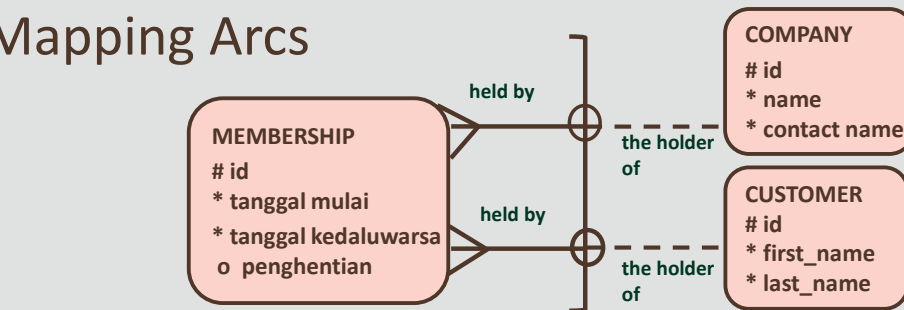
**CUBICLES (CBE)**

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | code        |
|          | *           | description |

## Mapping Arcs

- The entity that has the arc will map to a table that contains foreign keys from the tables on the “one” end of the relationships
- Note that even if the relationships in the arc are mandatory on the many side, the resulting foreign keys have to be optional (because one of them will always be blank). A check constraint stored in the database can easily do this

# Mapping Arcs



MEMBERSHIPS (MBP)

| Key Type | Optionality | Column name     |
|----------|-------------|-----------------|
| pk       | *           | id              |
|          | *           | start_date      |
|          | *           | expiration_date |
|          | o           | termination     |
| fk1      | o           | cpe_id          |
| fk2      | o           | cms_id          |

COMPANIES (CPE)

| Key Type | Optionality | Column name  |
|----------|-------------|--------------|
| pk       | *           | id           |
|          | *           | name         |
|          | *           | contact_name |

CUSTOMERS (CMS)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |
|          | *           | first_name  |
|          | *           | last_name   |

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Since the arc represents exclusive relationships, additional code is needed to enforce that only one of the foreign keys has a value for every row in the table.

A check constraint stored in the database can easily do this.

In the example, the code for the check constraint would look like this:

- CHECK (cpe\_id is not null AND cms\_id is null)
- OR (cpe\_id is null AND cms\_id is not null)

If the relationships were fully optional, you would add: OR (cpe\_id is null AND cms\_id is null)



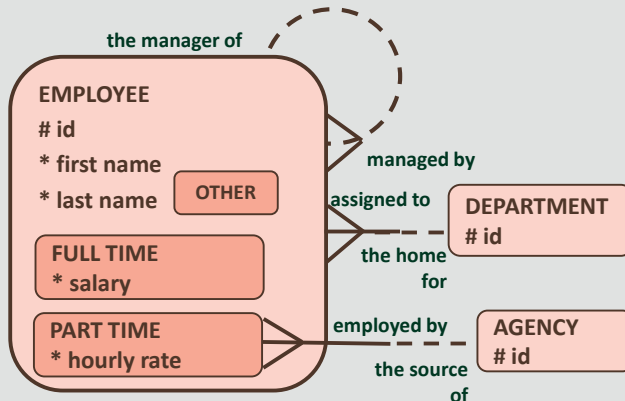
# Mapping Supertype/Subtypes

- Supertype/subtype entities can be mapped in multiple ways :
  - Single table implementation : one table is created regardless of number of subtypes, used when most of the attributes and relationships are shared and therefore at the supertype level
  - Two table implementation : a table is created for each of the subtypes (so there can be more than two tables), used when subtypes have little in common and few shared attributes and relationships

## Single Table Implementation

- The single table gets one column for each attribute of the supertype, along with the original optionality of the attribute
- The table also gets a column for each attribute belonging to the subtype, but the columns all become optional
- Additionally, a mandatory column should be created to act as a discriminator column to distinguish between the different subtypes of the entity

# Single Table Implementation



DEPARTMENTS (DPT)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |

AGENCIES (AGY)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |

EMPLOYEES (EPE)

| Key Type | Optionality | Column name |
|----------|-------------|-------------|
| pk       | *           | id          |
|          | *           | first_name  |
|          | *           | last_name   |
|          | o           | salary      |
|          | o           | hourly_rate |
| fk1      | *           | dpt_id      |
| fk2      | o           | agy_id      |
|          | *           | epe_type    |
| fk3      | o           | mgr_id      |

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**Identifiers:** Unique identifiers transform into primary and unique keys.

**Relationships:** Relationships at the supertype level transform as usual. Relationships at the subtype level are implemented as optional foreign-key columns.

**Integrity constraints:** A check constraint is needed to ensure that for each particular subtype, all columns that come from mandatory attributes are not null.

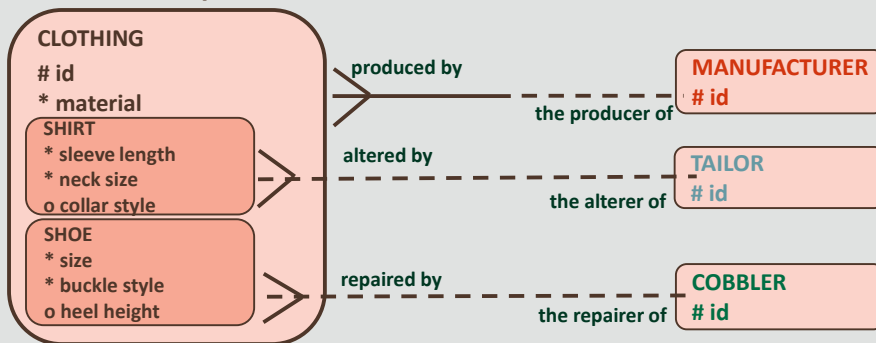
In the logical model, salary is mandatory for full-time employees and hourly rate is mandatory for part-time employees.

- When the EMPLOYEE supertype is implemented as a single table in the physical model, these attributes become optional.
- A check constraint is needed to enforce the business rules modeled in the ERD.

## Two Table Implementation

- One table per first-level subtype
- Each table gets one column for each attribute of the supertype along with its original optionality
- Each table also gets one column for each attribute belonging to the subtype along with its original optionality

# Two Table Implementation



SHIRTS (SHT)

| Key Type | Optionality | Column name   |
|----------|-------------|---------------|
| pk       | *           | id            |
|          | *           | material      |
|          | *           | sleeve_length |
|          | *           | neck_size     |
|          | o           | collar_style  |
| fk1      | o           | tlr_id        |
| fk2      | *           | mnr_id        |

SHOES (SHE)

| Key Type | Optionality | Column name  |
|----------|-------------|--------------|
| pk       | *           | id           |
|          | *           | material     |
|          | *           | size         |
|          | *           | buckle_style |
|          | o           | heel_height  |
| fk1      | o           | clr_id       |
| fk2      | *           | mnr_id       |

refers to **tailors**

refers to **manufacturers**

refers to **cobblers**

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**Identifiers:** The primary UID at the supertype level creates a primary key for each table. Secondary UIDs of the supertype become unique keys in each table.

**Relationships:**

- All tables get a foreign key for a relationship at the supertype level, with the original optionality (manufacturers)
- For relationships at the subtype levels, the foreign key is implemented in the table it is mapped to, original optionality is retained. (tailors, cobblers)

# Project Exercise

- DFo\_3\_4\_Project
  - Oracle Baseball League Store Database
  - Apply the Rules of Relationship Mapping to Transform Relationships



# Summary

- In this lesson, you should have learned how to:
  - Apply terminology mapping between the logical and physical models
  - Understand and apply the Oracle naming conventions for tables and columns used in physical models
  - Apply the rules of relationship mapping to correctly transform relationships



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